Mentoring Plan

FINESST companion document (2-page limit). ANONYMIZED — "the FI" and "the PI" only, no names or institutional identifiers. Items in [brackets] need the PI's confirmation or the institution's specifics — review this draft WITH the advisor before submission; the commitments in it are the PI's to make. Check whether the institution has a required mentoring-plan template before using this one.

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1. Purpose and roles

The FI is the intellectual lead of the proposed research: the FI designed the project, authored this proposal, and will execute the analyses, write the papers as first author, and release the data products. The PI serves as research mentor and PI of record: scientific guidance, domain expertise, institutional accountability, and career development. This plan states how that relationship operates for the award's duration — a designed structure, not an assumption.

2. Mentoring structure and cadence

- Weekly one-on-one meetings ([day/frequency to confirm]) between FI and PI: progress against the timeline, technical blockers, and interpretation of results. The FI sets the agenda; the PI probes assumptions and connects results to the wider literature.
- [Weekly/biweekly] research-group meetings: the FI presents work-in-progress to the group at least [once per month], building the habit of defending methods and results before an informed audience.
- Semester milestone reviews: at the start of each semester, FI and PI revisit the project timeline (§6 of the proposal) against actual progress, adjust scope where needed, and record the decisions. The honest-assessment section of the proposal (fallback plans per risk) is the standing agenda for these reviews.
- Committee alignment: the FI's dissertation committee reviews project progress at the institution's required milestones ([candidacy/qualifying exam, annual review, proposal defense — confirm which apply and when]), keeping the FINESST work and the degree requirements on one track.

3. What the mentoring specifically develops

The proposal names two genuine gaps in the FI's preparation; the mentoring plan is how they close.

Domain knowledge — Antarctic geomorphology and hydrology. The FI's methods background is strong but was built on temperate-landscape problems. The PI will direct a structured reading program in cold-desert process geomorphology (energy-limited melt, permafrost/active-layer dynamics, closed-basin lake behavior), [and/or supervise a directed-reading course in term X]. By the end of Year 1, the FI can defend the physical interpretation of every O1 driver to a polar audience, not just the statistics.

Statistical methodology — hierarchical modeling. The FI will complete graduate coursework in Bayesian/hierarchical statistical modeling ([course number/term to confirm]) before the Year-1 attribution analysis that depends on it. The PI will review model specifications at the design stage, before fitting — the point where review prevents mistakes instead of discovering them.

4. Professional development

- Conferences: the FI presents at AGU (cryosphere section) in [Year 1 or 2] and at a SCAR/ISAES Antarctic venue in [Year 2 or 3]. First posters, then a talk as results mature. The PI introduces the FI to the McMurdo Dry Valleys research community — including the LTER network whose data this project uses — at these venues.
- Publication: three first-author papers are planned (proposal §6 timeline). The PI's role is manuscript review and journal strategy; the FI drafts, submits, and handles review responses, with the PI advising. Authorship follows [institutional / disciplinary] norms, stated in advance: the FI is first author on the project's papers.
- Proposal craft: this proposal is itself a training artifact. During the award, the FI will draft at least one follow-on funding proposal ([e.g., a small analysis grant or observing/data proposal]) with PI review — building the skill the fellowship exists to develop: becoming a fundable independent investigator.
- Open science practice: the FI maintains the public code repository, documentation, and data releases described in the OSDMP; the PI reviews each public release. This is treated as a professional skill under active mentorship, not an afterthought.
- Teaching/outreach: [optional — add the institution's or FI's plans, e.g., one guest lecture per year, mentoring an undergraduate on a bounded sub-task.]

5. Career development

- The FI and PI complete an Individual Development Plan (IDP) in the first semester of the award and revisit it annually: target career path ([research faculty / agency scientist / industry — FI to state]), the skills and network that path needs, and which of them each project year should supply.
- The PI commits to timely, candid letters and introductions as the FI approaches graduation, and to structuring Year 3 so that dissertation completion and the project's synthesis deliverables are the same work, not competing work.

6. Communication, feedback, and safeguards

- Feedback: the PI provides written feedback on manuscripts and the annual NASA progress report within [two weeks] of receiving drafts. The FI flags blockers early rather than at milestone reviews; the weekly meeting exists so that no problem is more than a week old before the PI knows about it.
- Disagreement: scientific disagreements are resolved by evidence and recorded in the project's decision log. If the mentoring relationship itself needs mediation, the FI's dissertation committee and the department's graduate advisor are the named escalation path, per [institutional policy].
- Availability: if the PI is on leave or travel for an extended period, [named arrangement — e.g., a designated committee member] provides interim technical guidance so the FI is never unsupervised for more than [interval].

7. Assessment of the mentoring itself

Once per year, FI and PI explicitly review this plan — what mentoring worked, what was missing, what the coming year's emphasis should be — and revise it. The plan is a working document, treated as one.